Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-BH-A18J · Group B

Report date 31 Oct 2026

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**56**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**EUR**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**—**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-BH-A18J-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **2.5** · MSIsensor score: **0.08** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

99th

High risk tier · fused neuro-symbolic model



Probability of progression

64% by 36 mo

12 mo 19% · 24 mo 42% · 60 mo 88%

Model-estimated progression time

≈ 43 months

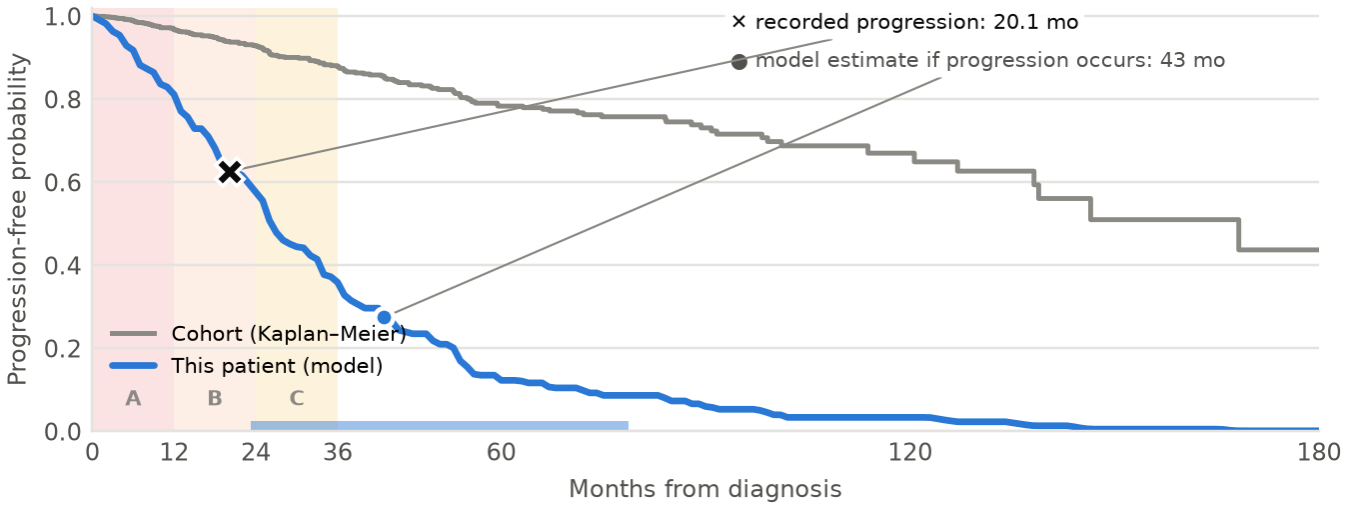
Window > 36 months · IQR 23–79 mo if progression occurs

Recorded progression

20.1 months

Group B · Progression in year 2

✗ recorded window differs · |error| 23 mo



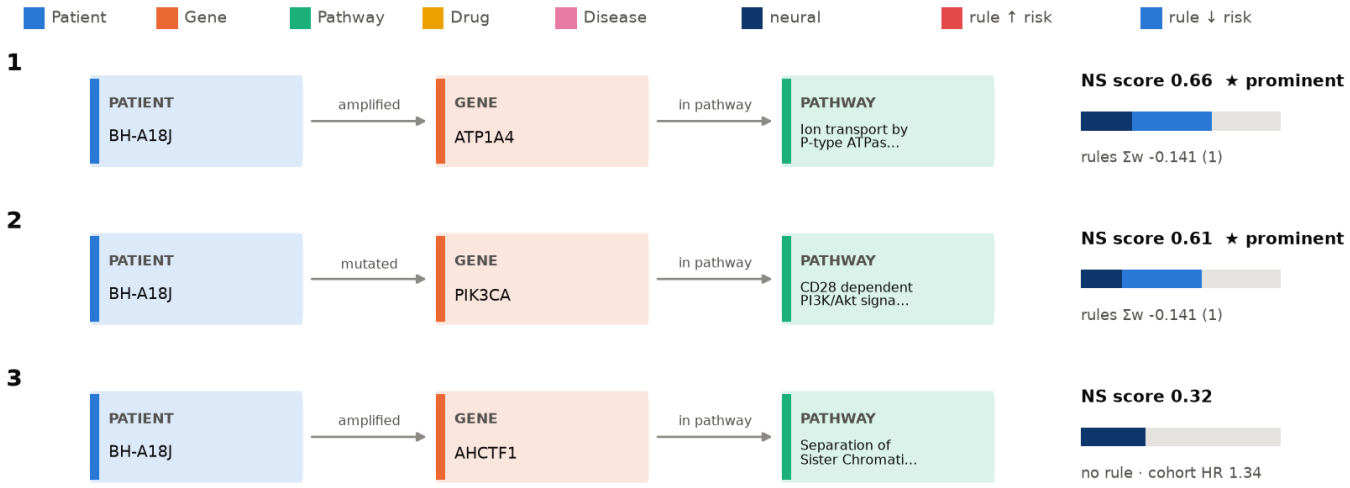
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 56-year-old patient has a LumA tumour, AJCC stage IV (T4, N2, M1). The fused neuro-symbolic model places them in the **99th risk percentile** (high tier). The strongest learned formula that holds for them (*Altered(p, g)* ∧ *InPathway(g, 'VEGFA-VEGFR2 Pathway')* ∧ *Altered(p, g)* ∧ *InPathway(g, 'Ion homeostasis')* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.141 per SD). The highest-scoring knowledge-graph path runs from **ATP1A4** to **Ion transport by P-type ATPases**; the learned formulas it instantiates point to slower progression (Σw -0.141). Progression was recorded at 20.1 months; the model's estimate given progression was 43 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

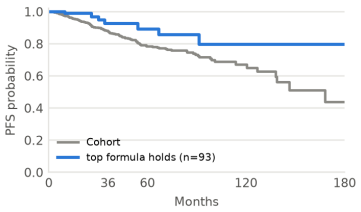


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-BH-A18J** amplified → **ATP1A4** in pathway → **Ion transport by P-type ATPases**	43	↓ **slower progression** · Σw -0.141 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'VEGFA-VEGFR2 Pathway') ∧ Altered(p, g) ∧ InPathway(g, 'Ion homeostasis') ⇒ Progress(p) cohort: HR 1.14, n = 94 carriers	**0.66**
2 ★	**TCGA-BH-A18J** mutated → **PIK3CA** in pathway → **CD28 dependent PI3K/Akt signaling**	34	↓ **slower progression** · Σw -0.141 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'VEGFA-VEGFR2 Pathway') ∧ Altered(p, g) ∧ InPathway(g, 'Ion homeostasis') ⇒ Progress(p) cohort: HR 0.98, n = 401 carriers	**0.61**
3	**TCGA-BH-A18J** amplified → **AHCTF1** in pathway → **Separation of Sister Chromatids**	54	No learned formula fires on this path; its score is neural only. cohort: HR 1.34, n = 211 carriers	**0.32**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

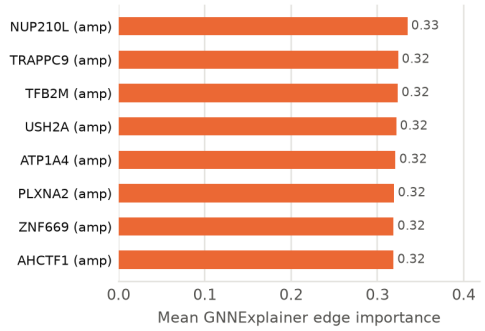
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.141 ↓ risk	Altered(p, g) ∧ InPathway(g, 'VEGFA-VEGFR2 Pathway') ∧ Altered(p, g) ∧ InPathway(g, 'Ion homeostasis') ⇒ Progress(p) [KG]	RL-proposed	93 pts
+0.134 ↑ risk	TStage(p, T4) ⇒ Progress(p)	single atom	27 pts
+0.018 ↑ risk	Amplified(p, GOLPH3L) ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Complement cascade') ⇒ Progress(p) [KG]	mined pair	42 pts
-0.016 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	408 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
KMT2C · driver	Mutation	p.Q318E	4	0.32
PIK3CA · driver	Mutation	p.E545K	70	0.31
NUP210L	High-level amplification	GISTIC ±2	0	0.33
TRAPPC9	High-level amplification	GISTIC ±2	2	0.32
TFB2M	High-level amplification	GISTIC ±2	2	0.32
USH2A	High-level amplification	GISTIC ±2	0	0.32
ATP1A4	High-level amplification	GISTIC ±2	4	0.32
PLXNA2	High-level amplification	GISTIC ±2	4	0.32
ZNF669	High-level amplification	GISTIC ±2	1	0.32
AHCTF1	High-level amplification	GISTIC ±2	7	0.32
ZNF124	High-level amplification	GISTIC ±2	1	0.32
FAM72C	High-level amplification	GISTIC ±2	0	0.32



40 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); ATP1A4 → Rubidium Rb-82 (DB09479); PARP1 → Theophylline (DB00277); PARP1 → Zinc (DB01593) … and 22 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0CS	73	IV	LumA	0.61	no progression, followed 77.2 mo
TCGA-AN-A0XP	69	III	LumA	0.60	no progression, followed 0.3 mo
TCGA-A8-A09B	58	III	LumA	0.56	no progression, followed 12.0 mo
TCGA-A8-A08T	64	IV	LumA	0.54	progressed at 84.1 mo (group D)
TCGA-AO-A12A	47	II	LumA	0.52	no progression, followed 102.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.